Darija charGPT Learning Roadmap

A beginner-friendly path to build and train a tiny GPT-style model in C++ that learns from Darija text and can chat in the terminal.

0. Setup	1. Dataset	2. Tokenizer	3. Bigram	4. Tiny NN	5. Mini GPT	6. Train	7. Chat	8. Save/Load	9. CUDA later

Main roadmap

0	**Setup** • Learn/use: C++, g++, CMake, Git, basic terminal. • Create the project folders. • Start with CPU only. Do not use CUDA yet.
1	**Prepare the dataset** • Use Darija Open Dataset (DODa). • Extract Darija sentences. • Lowercase, remove empty lines, keep Latin Darija first. • Save clean text to data/darija.txt.
2	**Character tokenizer** • Convert text to numbers and numbers back to text. • Build char_to_id, id_to_char, encode(), decode(), and vocab_size. • Checkpoint: "salamo 3likom" -> ids -> "salamo 3likom".
3	**Bigram model** • Learn the idea: current character -> next character. • Train a simple table before building GPT. • Goal: generate Darija-like random text and see loss go down.
4	**Tiny neural network** • Learn embeddings, matrix multiplication, softmax, cross-entropy, gradient descent. • Architecture: character id -> embedding -> linear layer -> softmax. • Goal: predict next characters better than random.
5	**Mini GPT / Transformer** • Add token embeddings, position embeddings, attention, causal mask, feed-forward, LayerNorm, residuals. • Start small: context=64, embedding=64, heads=2, layers=1-2, batch=16. • Checkpoint: training loss decreases.
6	**Train on Darija** • Load text, encode it, create batches, forward pass, loss, backprop, update weights. • Print loss every few steps. • Good first target: loss below 3.0; better: around 2.0-2.5.
7	**Add chat mode** • Use prompt format: User: ... / Bot: • Model continues generating after Bot:. • Fine-tune on data/chat_darija.txt to improve chat behavior.
8	**Save and load model** • Save model weights to models/darija_gpt.bin. • Save vocabulary/tokenizer too. • Checkpoint: close program, reopen, load model, and chat.

9

CUDA later

• Only after CPU version works.
• Add matrix multiplication kernel, softmax kernel, attention kernel, batching optimization.
• CUDA before understanding the model will make the project much harder.

Project structure

Create this folder layout:

```
darija-charGPT/
■■■ data/
■■■ scripts/
■■■ src/
■■■ build/
■■■ models/
■■■ README.md
```

Dataset plan

Your first goal is to create one clean file: **data/darija.txt**. Later, add **data/chat_darija.txt** for chat behavior.

Example data/darija.txt:

```
salam labas 3lik
kifach nta
wach t9dr t3awni
bghit nt3lm l artificial intelligence
chno smitk
```

Example data/chat_darija.txt:

```
User: salam
Bot: wa 3alaykom salam, labas?

User: kifach nta?
Bot: labas, nta?

User: chno smitk?
Bot: smiti DarijaGPT.
```

Core build flow

Dataset	Tokenizer	Batch Loader	GPT Model	Loss	Backprop	Optimizer	Save Model	Generate Text

Flow in words: GitHub DODa repo -> extract Darija sentences -> clean text -> save to data/darija.txt -> train char-level GPT -> generate Darija text -> add chat prompt format.

Files to build by phase

Phase	Files
Tokenizer	src/tokenizer.hpp, src/tokenizer.cpp, src/test_tokenizer.cpp
Bigram	src/bigram.hpp, src/train_bigram.cpp, src/generate_bigram.cpp
Tiny NN	src/matrix.hpp, src/nn.hpp, src/train_nn.cpp
Mini GPT	src/model.hpp, src/attention.hpp, src/transformer_block.hpp, src/layernorm.hpp, src/optimizer.hpp, src/train_gpt.cpp
Chat	src/chat.cpp
CUDA later	src/cuda/matmul.cu, src/cuda/softmax.cu, src/cuda/attention.cu

30-day practical roadmap

Week 1 - C++ + dataset + tokenizer

• Read file
• Clean text
• Build vocabulary
• Encode/decode

Goal: "salamo 3likom" -> ids -> "salamo 3likom"

Week 2 - Bigram model

• Bigram table
• Sampling
• Loss calculation
• Simple generation

Goal: generate Darija-like random text.

Week 3 - Tiny neural network

• Matrix class
• Embeddings
• Linear layer
• Softmax
• Training loop

Goal: loss goes down.

Week 4 - Mini GPT

• Attention
• Causal mask
• Transformer block
• Generation
• Chat mode
• Save/load

Goal: terminal Darija chatbot.

Recommended learning order

C++ -> Tokenizer -> Bigram -> Neural network basics -> Embeddings -> Attention -> Transformer block -> Training loop -> Text generation -> Chat mode -> CUDA

Final target version

darija-charGPT v0.1

• CPU only
• Character-level tokenizer
• Small GPT
• Trained on DODa Darija text
• Fine-tuned on small chat examples
• Terminal chat
• Save/load model

Main mindset

Do not aim for perfect answers. Aim for understanding: tokenizer, training, loss, attention, generation, and a tiny Darija chatbot you can run in the terminal.

Visual roadmap